Remote Control of the Motion of a Differential Drive Robot by Moving the Palm of the Hand

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Agenda

1. Introduction
2. Objectives
3. System Design
4. Implementation
5. Results
6. Conclusion

Proposed Project

- A wireless hand gesture control system for a differential drive robot.
- The robot motion is controlled by palm movement using the MPU6050 motion sensor.
- Commands are transmitted wirelessly through Bluetooth communication.
- The system provides intuitive and real-time robot navigation.



Problem Statement & Importance

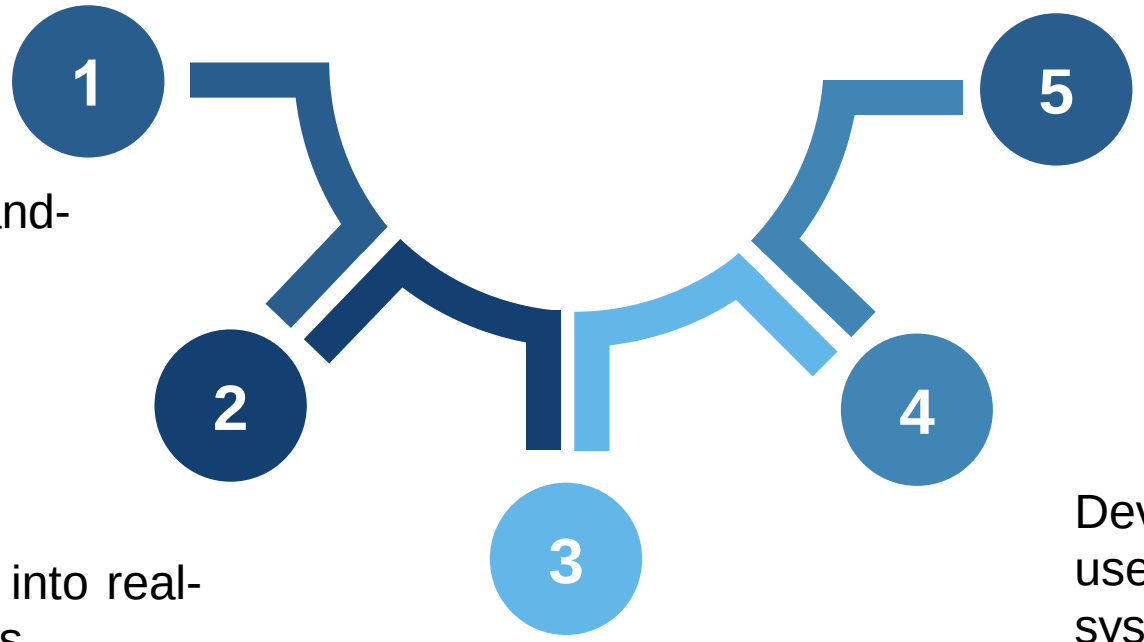
Problem Statement

- Elderly individuals and people with disabilities often face difficulties when using conventional control systems for robots and automated devices.
- Existing human–robot interaction methods are frequently complex, non-intuitive, and require physical effort.
- The limited accessibility of current systems reduces the practical use of assistive robots in daily life.
- There is a growing need for simple, natural, and user-friendly control mechanisms.

Importance of the Project

- The project introduces an intuitive robot control system based on natural hand gestures.
- The system integrates wearable sensing, wireless communication, and embedded processing technologies.
- It simplifies human–robot interaction and reduces the learning curve for users.
- The project demonstrates practical applications of sensor technologies and embedded systems in assistive robotics.
- It also provides a foundation for future developments in gesture-based control and human–robot collaboration.

Project Objectives



1. Design and implement a hand-gesture-controlled mobile robot.
2. Translate hand movements into real-time robot motion commands.
3. Establish reliable wireless communication using Bluetooth modules.
4. Develop an intuitive and user-friendly control system for elderly and disabled users.
5. Create a scalable platform for future assistive robotics applications.

Related Work

Study	Technology Used	Strengths	Limitations	Contribution to Our Project
Vala, 2018	ADXL335 accelerometer + Arduino Uno + RF communication	Simple gesture mapping and lightweight implementation	RF communication is less reliable and more sensitive to interference	Inspired the transmitter–receiver architecture and gesture mapping approach
Rohan et al., 2024	MPU-6050 sensor + Camera + CNN + MQTT	Accurate real-time gesture recognition and scalable architecture	High computational requirements and sensitivity to lighting conditions	Supported the use of MPU-6050 for reliable real-time gesture control
Vision-Based Gesture Systems	Camera and image processing algorithms	High gesture recognition accuracy under controlled conditions	Complex processing and environmental sensitivity	Reinforced the decision to use lightweight wearable sensor-based control

Theoretical Background & Mathematical Modeling

MPU-6050 Gesture Detection

- Measures hand tilt using acceleration readings along X and Y axes.
- Converts palm movement into estimated tilt angles θx and θy.
- A threshold of approximately ±45° is used to detect intentional gestures.

Threshold-Based Control Logic

- $\theta y > 45°$ → Forward command
- $\theta y < -45°$ → Backward command
- $\theta x > 45°$ → Right turn command
- $\theta x < -45°$ → Left turn command
- Neutral position → Stop command

Differential Drive Model

- The robot uses two driving wheels and one free caster wheel.
- Equal wheel speeds produce forward/backward motion.
- Different wheel speeds generate turning motion.

Main Motion Equations

$$V = (VR + VL) / 2$$

$$\omega = (VR - VL) / L$$

- Where:

  V = robot linear velocity

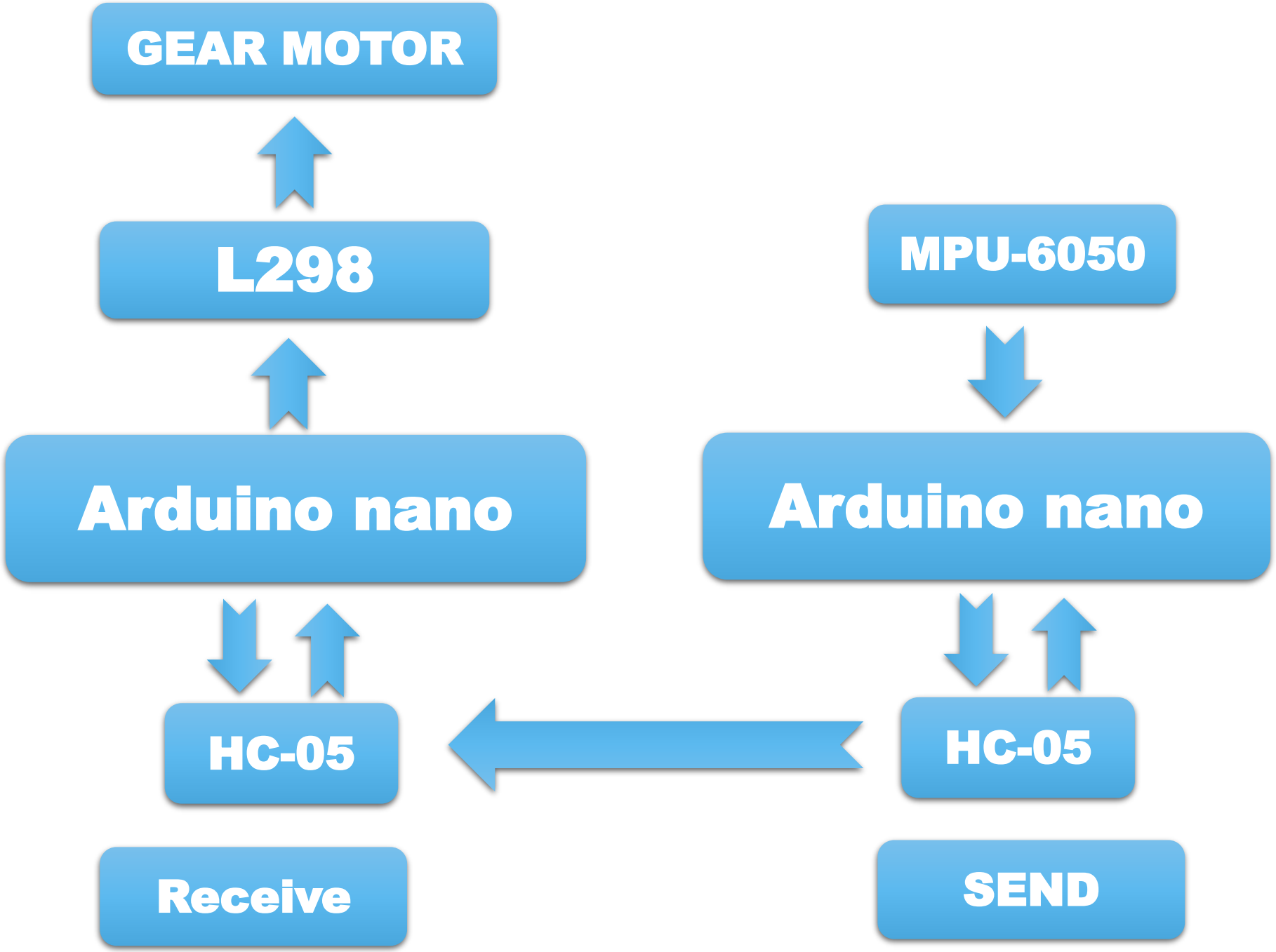
  ω = robot angular velocity

  VR, VL = right and left wheel velocities

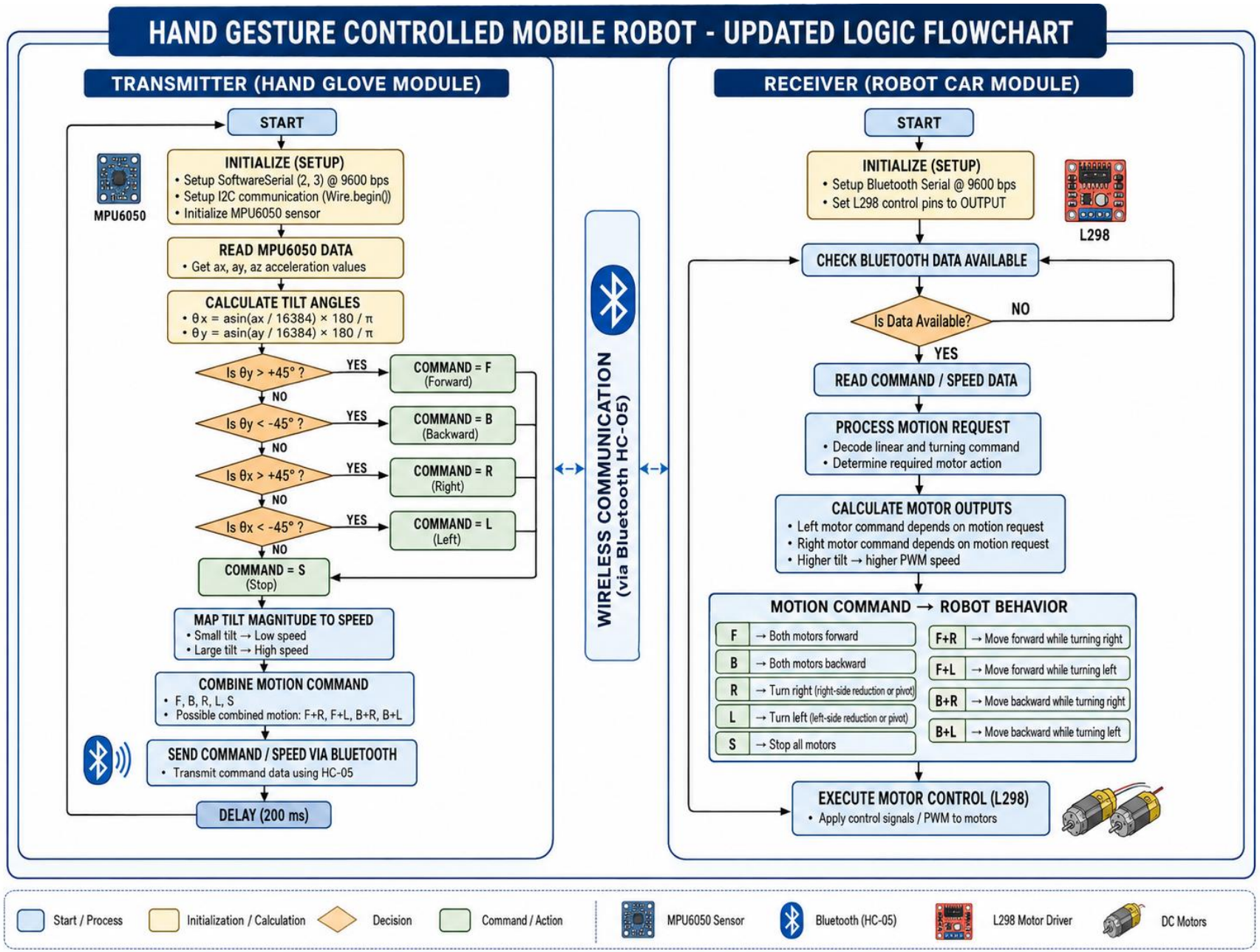
  L = distance between the driving wheels

System Architecture

Block Diagram



Flowchart


HAND GESTURE CONTROLLED MOBILE ROBOT - UPDATED LOGIC FLOWCHART
TRANSMITTER (HAND GLOVE MODULE)
START
MPU6050
INITIALIZE (SETUP)
• Setup SoftwareSerial (2, 3) @ 9600 bps
• Setup I2C communication (Wire.begin())
• Initialize MPU6050 sensor
READ MPU6050 DATA
• Get ax, ay, az acceleration values
CALCULATE TILT ANGLES
• θx = asin(ax / 16384) × 180 / π
• θy = asin(ay / 16384) × 180 / π
Is θy > +45° ?
YES
COMMAND = F (Forward)
NO
Is θy < -45° ?
YES
COMMAND = B (Backward)
NO
Is θx > +45° ?
YES
COMMAND = R (Right)
NO
Is θx < -45° ?
YES
COMMAND = L (Left)
NO
COMMAND = S (Stop)
MAP TILT MAGNITUDE TO SPEED
• Small tilt → Low speed
• Large tilt → High speed
COMBINE MOTION COMMAND
• F, B, R, L, S
• Possible combined motion: F+R, F+L, B+R, B+L
SEND COMMAND / SPEED VIA BLUETOOTH
• Transmit command data using HC-05
DELAY (200 ms)
WIRELESS COMMUNICATION (via Bluetooth HC-05)
RECEIVER (ROBOT CAR MODULE)
START
INITIALIZE (SETUP)
• Setup Bluetooth Serial @ 9600 bps
• Set L298 control pins to OUTPUT
L298
CHECK BLUETOOTH DATA AVAILABLE
Is Data Available?
NO
YES
READ COMMAND / SPEED DATA
PROCESS MOTION REQUEST
• Decode linear and turning command
• Determine required motor action
CALCULATE MOTOR OUTPUTS
• Left motor command depends on motion request
• Right motor command depends on motion request
• Higher tilt → higher PWM speed
MOTION COMMAND → ROBOT BEHAVIOR
F → Both motors forward
B → Both motors backward
R → Turn right (right-side reduction or pivot)
L → Turn left (left-side reduction or pivot)
S → Stop all motors
F+R → Move forward while turning right
F+L → Move forward while turning left
B+R → Move backward while turning right
B+L → Move backward while turning left
EXECUTE MOTOR CONTROL (L298)
• Apply control signals / PWM to motors
Start / Process
Initialization / Calculation
Decision
Command / Action
MPU6050 Sensor
Bluetooth (HC-05)
L298 Motor Driver
DC Motors

System Technical Specifications and Detailed Design

Technical Specifications

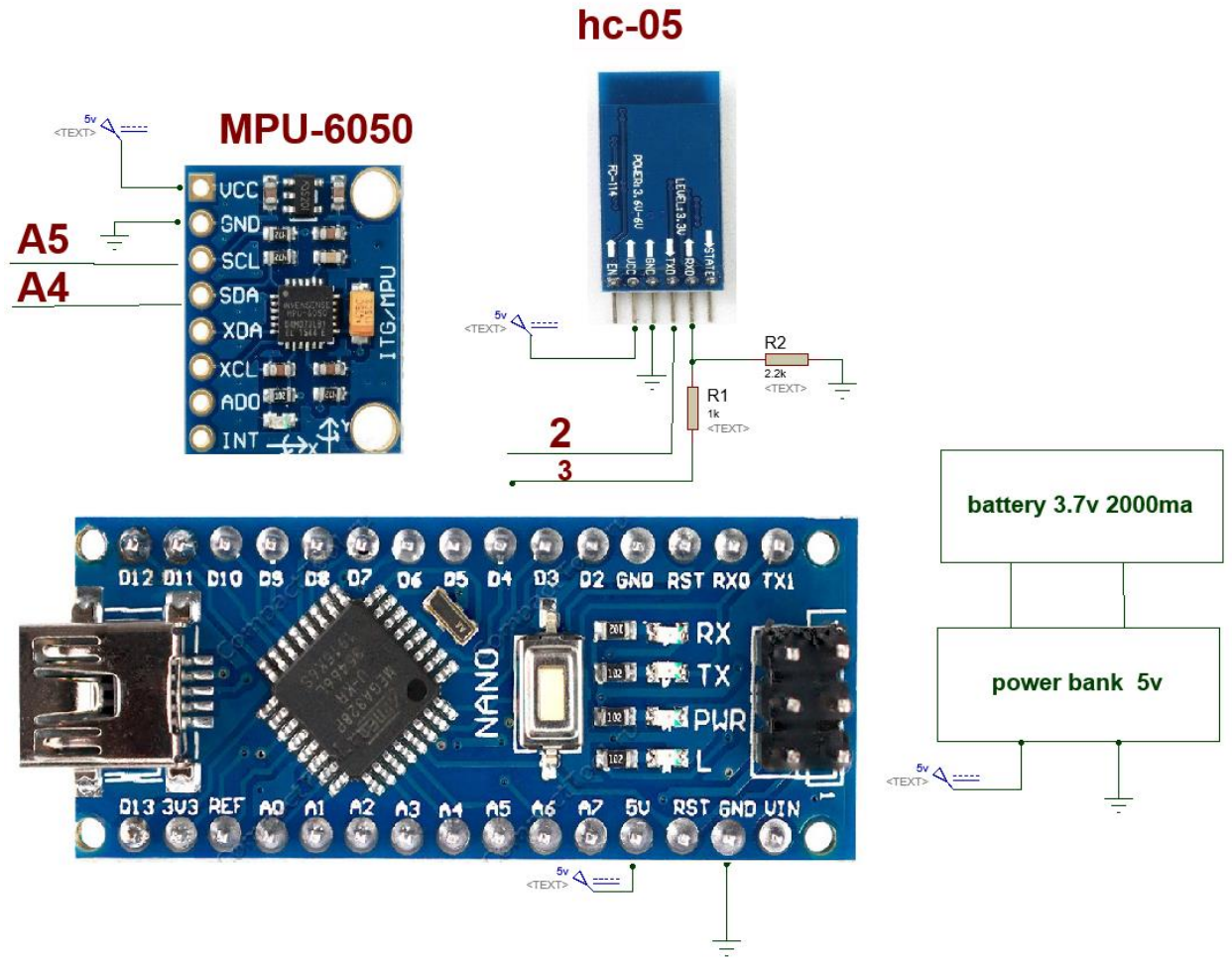
Category	Parameter	Specification
Weight	Robot Weight	≤ 200 g
Weight	Controller Weight	≤ 120 g
Speed	Maximum Speed	0.25 – 0.35 m/s
Operating Range	Wireless Range	Up to 10–30 m
Drive System	Robot Mobility	Differential Drive System
Motion Functions	Movement Modes	Forward, Backward, Left, Right, Stop
Sensing System	Motion Detection	MPU-6050 Motion Sensor
Power System	Power Source	Rechargeable Lithium Battery
Performance	Response Time	< 1 second
Performance	Operating Time	20–30 minutes
Communication	Wireless Link	HC-05 Bluetooth Connection

Detailed Design Components

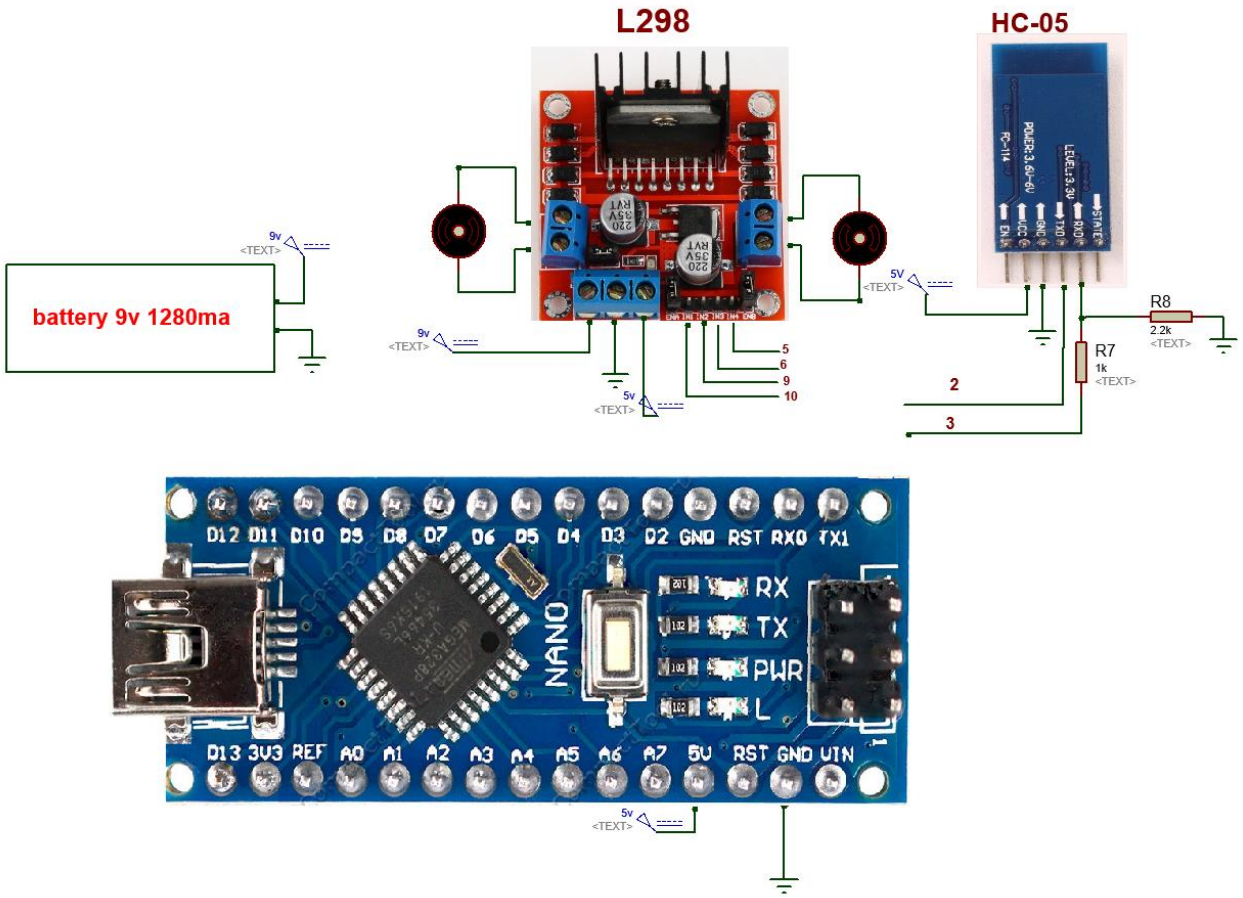
Component	Function
Arduino Nano	Processes sensor data and controls system operation
MPU-6050 Sensor	Detects hand tilt and motion
HC-05 Bluetooth Module	Provides wireless communication between transmitter and receiver
L298 Motor Driver	Controls the direction and operation of the DC motors
DC Gear Motors	Drive the robot wheels
Robot Car Kit / Chassis	Supports the robot structure and electronic components
Driving Wheels	Convert motor rotation into robot motion

Hardware Implementation

Transmitter diagram



Reception Diagram



Results & Discussion

Practical Testing Results

The robot successfully executed the five main commands:

- Forward
- Backward
- Right
- Left
- Stop

System Performance

- Overall success rate achieved: **88%**
- Response time ranged between **1–2 seconds**
- Bluetooth communication was stable during indoor testing
- Robot movement was generally stable on flat surfaces

Discussion

- Forward and backward commands achieved higher reliability than turning motions.
- Turning gestures required more stable hand positioning and sensor calibration.
- The system successfully demonstrated real-time gesture-based robot control using low-cost components.

Current Limitations

- No obstacle detection system
- Combined motion was not implemented
- Bluetooth operating range is limited

Conclusion

• A hand-gesture-controlled differential drive robot was successfully designed and implemented.

• The system enabled intuitive wireless robot control using the MPU-6050 sensor and Bluetooth communication.

• The implemented prototype successfully performed the main movement commands:

• Forward

• Backward

• Right

• Left

• Stop

• The robot achieved an overall success rate of 88% with acceptable response time and stable operation during testing.

• The project demonstrated the practical integration of embedded systems, wireless communication, and human–robot interaction concepts.

• The results confirm that low-cost gesture-control systems can provide effective and accessible robotic solutions for future assistive applications.

Future Work

- Implement obstacle detection and avoidance using ultrasonic sensors.
- Develop proportional speed control based on hand tilt angle.
- Support combined movements such as forward motion while turning.
- Improve gesture recognition accuracy through better sensor calibration and filtering.
- Extend the wireless communication range using Wi-Fi or IoT technologies.
- Integrate additional assistive modules such as a robotic arm or camera system.

Thank you for listening